

RV College of Engineering

Project Based Learning Report CAD CAM & Robotics – IM343AI

2025-26



INDUSTRY VISIT REPORT

STUDENT GROUP

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1. Overview & Objectives

As an integral component of the Experiential Learning (EL) framework for the CAD/CAM & Robotics (IM343AI) curriculum, the student team undertook structured industrial visits to three distinct manufacturing facilities located in the Peenya Industrial Area, Bengaluru, in June 2026. The core objective of these visits was to bridge the gap between classroom-taught theoretical concepts—such as numerical control (NC) part programming, geometric modeling, and flexible automation—and their physical, large-scale execution on the shop floor.

The manufacturing sector increasingly relies on the seamless integration of Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM). By observing operations ranging from precision machine tool building to heavy industrial equipment fabrication and automated sheet metal processing, this experiential study provided direct exposure to the realities of production engineering. The team interacted closely with industry experts, design engineers, and shop-floor supervisors to analyze how process parameters, machine utilization rates, quality control loops, and manufacturing layouts dictate plant productivity and operational efficiency.

2.Detailed Industry Profiles & Observations

Visit 1 — Sands Precision Pvt. Ltd.

Location: Peenya Industrial Area, Bengaluru

Industry Guide: Mr. Dinesh Karanth

Date: 10 June 2026

About the Company

Sands Precision Pvt. Ltd. Is a specialized niche manufacturer of Computer Numerical Control (CNC) machine tools, focusing primarily on the production of high-precision CNC turning centers and Vertical Machining Centers (VMCs). The enterprise serves high-accuracy downstream sectors including aerospace, automotive, defense, and general precision engineering. Operating with a lean workforce of approximately 30 personnel in a single-shift model, the facility outputs nearly 100 finished CNC machines annually. The plant operates at an average machine utilization rate of 45%, a metric characteristic of low-volume, highly customized machine tool assembly.

Technical Observations & Analysis

Under the guidance of Mr. Dinesh Karanth, the team tracked the entire value stream of a CNC machine tool from structural casting preparation to final dispatch.

The CNC Lifecycle: The production sequence begins with the inspection of heavy structural castings (beds and columns). These undergo precision machining to create the foundational reference surfaces. Sub-assembly phases involve the integration of linear motion (LM) guideways, ball screws, and spindle cartridges, culminating in the wiring of the electrical console and CNC controllers (such as Fanuc or Siemens units).

CAD/CAM Integration: CAD software is utilized internally for the structural design, finite element analysis (FEA) of machine frames, and component modeling. Concurrently, CAM software generates the toolpaths and multi-axis G-code required to machine the precision components that form the machine's internal architecture.

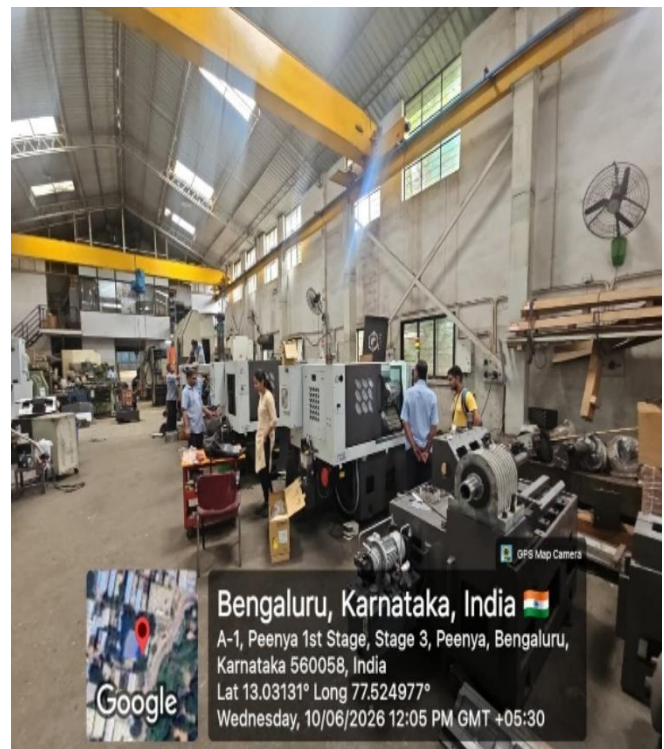
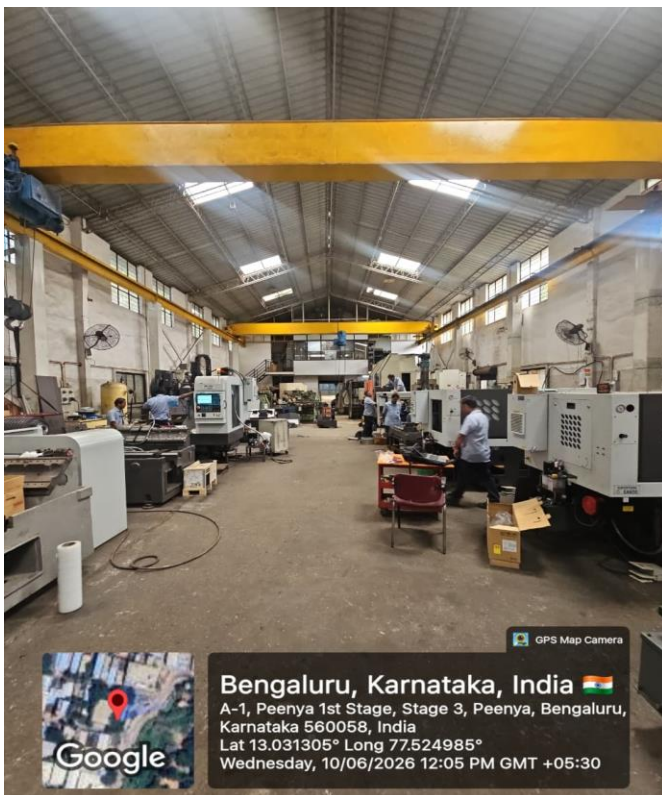
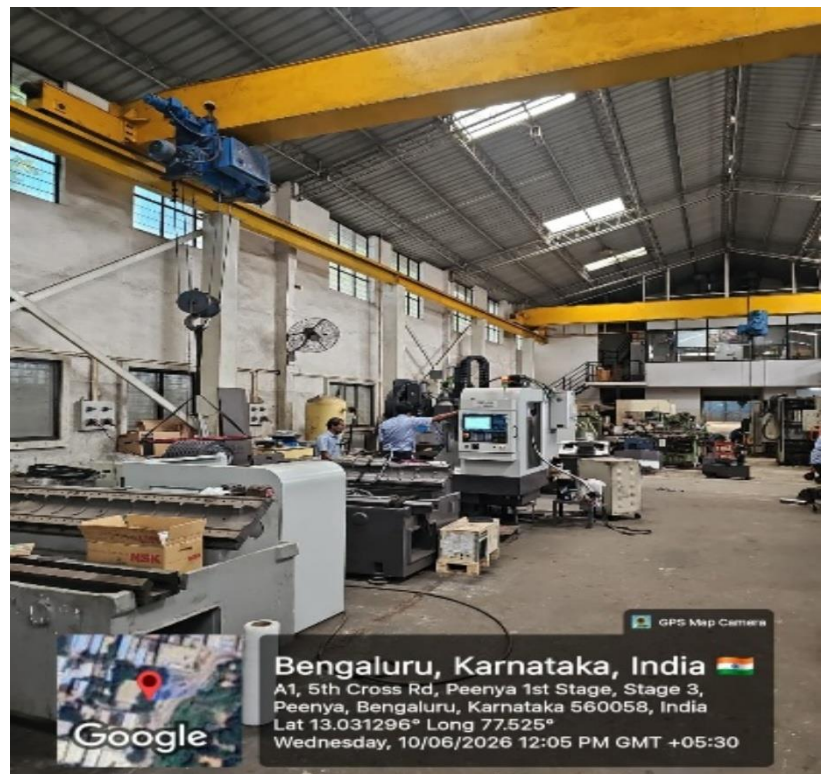
Metrology & Geometric Alignment: To guarantee sub-micron accuracy, quality inspection heavily relies on Coordinate Measuring Machines (CMMs) and electronic surface roughness testers. The physical assembly process demands meticulous scraping and alignment procedures to eliminate geometric errors (parallelism, perpendicularity, and runout) before the machine undergoes laser interferometry calibration.

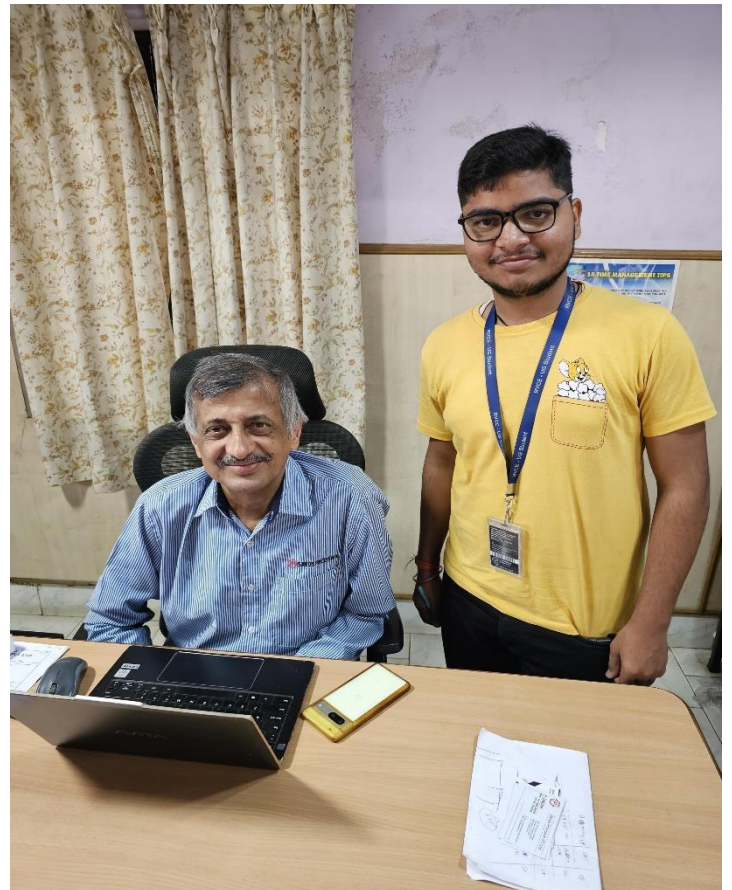
Capacity & Utilization Analytics: The 45% machine utilization rate was analyzed through an industrial engineering lens. Unlike continuous mass production, machine tool building is intrinsically a job-shop / batch production setup. Frequent pauses are required for high-precision manual alignments, custom engineering adjustments per client

specifications, and extensive iterative quality testing, which naturally limits continuous machine runtime.

Learning Outcome

Visiting Sands Precision provided a macro-level understanding of the manufacturing ecosystem. It demonstrated that the precision of hundreds of downstream factories depends entirely on the geometric accuracy built into these primary machine tools, reinforcing the absolute necessity of strict tolerances and robust metrological discipline.





Visit 2 — United Concord Industries

Location: Peenya Industrial Area, Bengaluru

Date: 10 June 2026

About the Company

United Concord Industries represents a large-scale, high-volume manufacturing paradigm, specializing in the engineering and production of heavy industrial pelletizing equipment. Their machinery is widely deployed in material processing industries, including plastics recycling, food processing, compounding, and chemical processing. The company scales its operations with over 250 personnel, maintaining a high machine utilization rate of approximately 85%. This reflects a continuous, demand-driven production schedule backed by structured workflow planning.

Technical Observations & Analysis

The visit focused on the workflow mechanics of handling large-scale structural assemblies and continuous manufacturing loops.

Shop Floor Layout & Material Flow: The facility utilizes a hybrid plant layout that combines cellular manufacturing for specialized sub-assemblies with a product-focused flow line for final machinery integration. Heavy machining areas feature an array of large-capacity

conventional and CNC equipment, including heavy-duty lathes, horizontal milling machines, and radial drilling machines operating in unison.

Complex Assembly Integration: Pelletizing machines comprise massive, intricate sub-systems requiring the precise marrying of heavy fabricated frameworks with precision-machined internal components (such as die plates, cutter hubs, and shafts) and outsourced components (motors, gearboxes, and pneumatic actuators).

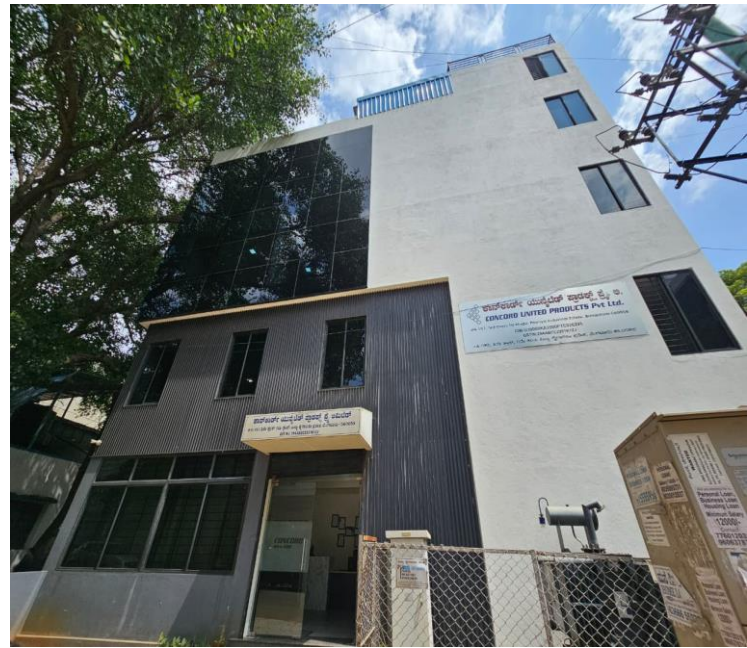
High-Utilization Drivers: The 85% utilization rate is achieved through aggressive production planning and scheduling, minimized setup times, and standardized component designs. The high volume of recurring orders allows the plant to run continuous shifts with optimal machine uptime.

Standardization & Lean Practices: CAD/CAM tools are central to enforcing component standardization. By creating parametric design libraries, the company ensures that replacement parts for field equipment can be rapidly programmed, machined, and dispatched. The shop floor clearly reflects lean manufacturing principles, featuring marked transport lanes, designated staging areas for Work-in-Progress (WIP), and structured material handling via overhead gantry cranes.

Learning Outcome

United Concord Industries served as an excellent case study for high-volume operational efficiency. The visit highlighted how robust production planning, component standardization, and optimized plant layouts are critical to sustaining high machine utilization and managing complex supply chains at scale.





Visit 3 — Sree Ranga Engineering Works

Location: Peenya Industrial Area, Bengaluru

Date: 10 June 2026

About the Company

Sree Ranga Engineering Works is a highly agile, technology-driven job-shop specializing in precision sheet metal fabrication and advanced CNC laser cutting services. Employing roughly 50 skilled workers, the enterprise provides rapid prototyping and component manufacturing services across diverse engineering sectors. The facility relies on a digital-to-physical workflow, utilizing advanced 3D CAD modeling software like SolidWorks and 2D drafting packages like AutoCAD to drive high-speed CNC laser cutters and automated press brakes.

Technical Observations & Analysis

The team observed the direct conversion of digital vector geometry into physical sheet metal components.

CNC Laser & Profiling Operations: The core of the facility features high-power CNC laser cutting systems that execute complex 2D profiling on varying gauges of sheet metal. The machine controllers interpret profile vectors directly from CAD data, converting them into precise X-Y coordinate toolpaths that regulate cutting speeds and assist-gas

pressures. This fast-paced digital execution directly mirrored the theoretical foundations of the student team's own academic projects.

CNC Bending & Part Nesting: Post-cutting, parts are transferred to CNC press brakes for precise bending. The bend sequences, tonnage calculations, and back-gauge positions are derived computationally from the unfolded 3D CAD models—mirroring the mathematical logic of standard G-code generation.

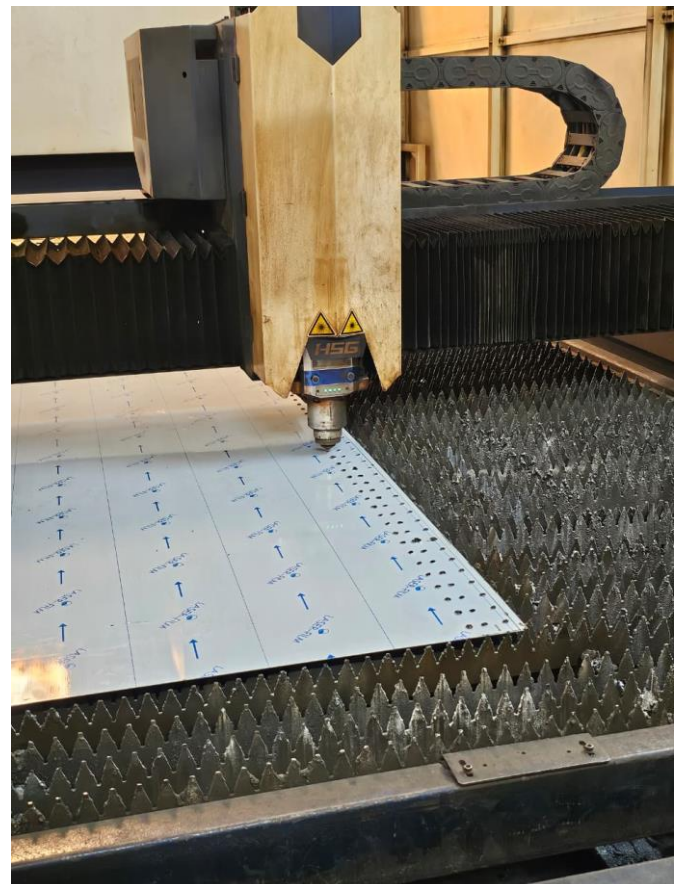
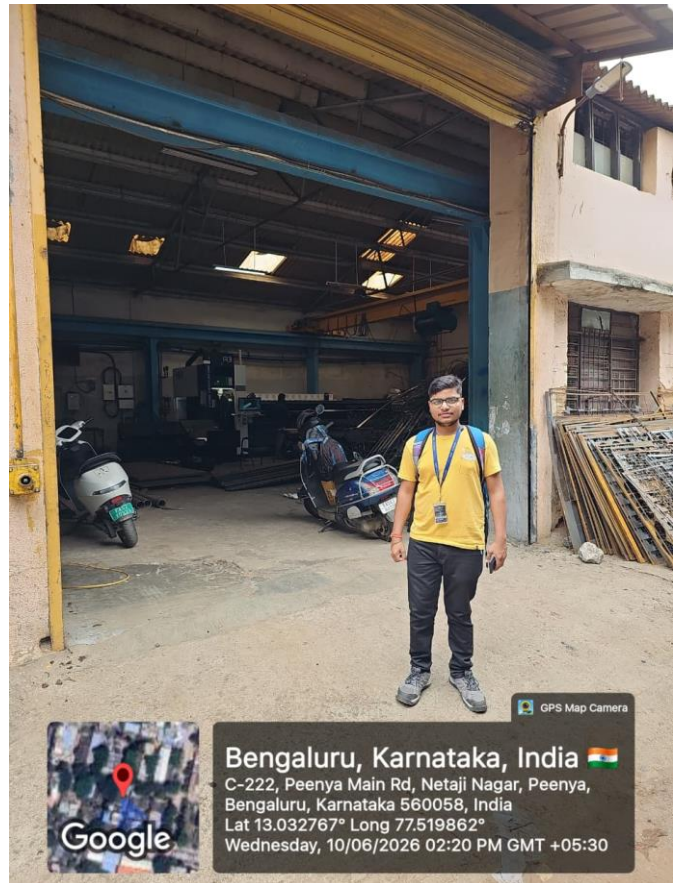
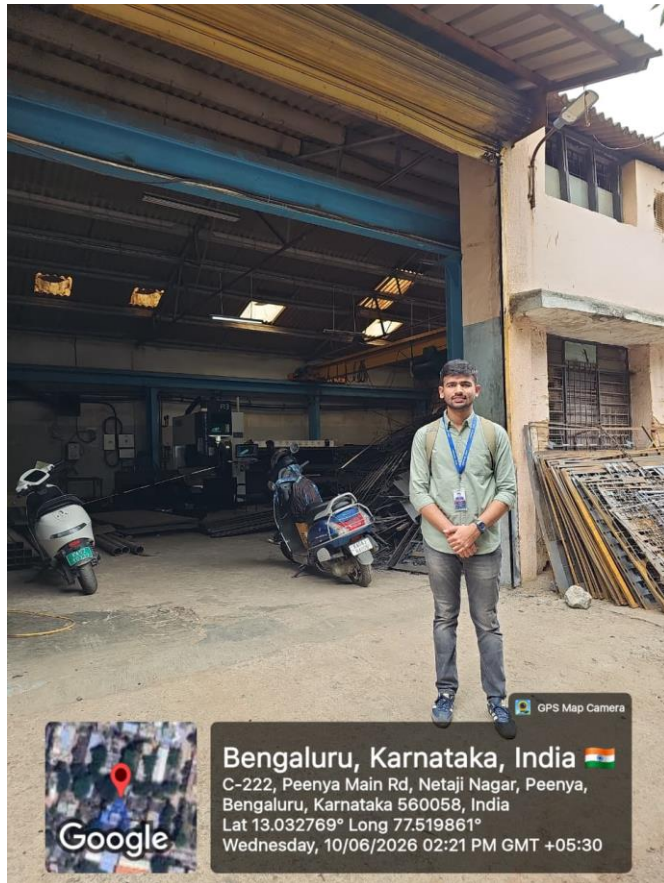
The Economics of Nesting: A major highlight was the automated nesting process executed via CAM software. Engineers arrange multiple part profiles onto a standard stock sheet of metal to optimize material yield. The team observed how even minor improvements in nesting efficiency directly reduce scrap rates, optimize material utilization, and protect profit margins on expensive raw materials.

Human-Machine Collaboration: While the initial blanking and forming processes are heavily automated via CNC systems, subsequent stages—such as jig-based welding, grinding, surface treatment, and manual assembly—rely on skilled labor, illustrating a balanced manufacturing cell ecosystem.

Quality Assurance: Dimensional verification on the floor is executed rapidly using precise physical instruments including digital vernier calipers, precision squares, and custom angle gauges to cross-reference physical bends with drawing specifications.

Learning Outcome

Sree Ranga Engineering Works showcased the high versatility, speed, and material efficiency of CAD/CAM software within the sheet metal industry. It provided practical validation of how component nesting and automated programming minimize lead times and material waste in an agile job-shop environment.



3. Consolidated Industrial Learnings

An analysis of all three distinct manufacturing environments reveals several overarching industrial principles:

The Automation-Labor Continuum: Across all facilities, automation (CNC, lasers, PLCs) is not used to completely replace human labor, but rather to enhance process capability, repeatability, and throughput. Skilled human operators remain crucial for high-level part programming, machine setups, real-time quality loops, and proactive maintenance.

CAD/CAM as an Operational Backbone: Computer-Aided Design and Manufacturing is no longer limited to the engineering design office. It serves as the primary data highway of the modern factory, connecting the initial digital concept directly to the toolpaths, nesting routines, and axis movements of the shop floor machinery.

Process Discipline Over Machinery: High-end machinery is only as effective as the processes surrounding it. Factors such as rigid workholding design, optimal cutting tool selection, systematic process planning, and structured metrology loops are what ultimately dictate final product quality.

Scalability of Fundamental Principles: Whether observing a boutique 30-person machine builder or a large 250-person heavy equipment plant, the foundational principles of industrial engineering—including cycle time reduction, layout optimization, capacity planning, and tolerance stacks—remain identical.

Socio-Technical Synergies: Beyond technical acumen in operating CAD/CAM systems, modern manufacturing environments demand strong communication skills, collaborative problem-solving, and cross-functional agility to successfully navigate the pipeline from design to delivery.

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- Industrial Site Study: Sands Precision Pvt. Ltd., Peenya Industrial Area, Bengaluru. (May 2025). Direct technical interactions and shop-floor brief by Mr. Dinesh Karanth.
- Industrial Site Study: United Concord Industries, Peenya Industrial Area, Bengaluru. (May 2025). Plant tour and process workflow briefings with plant engineers and supervisors.
- Industrial Site Study: Sree Ranga Engineering Works, Peenya Industrial Area, Bengaluru. (May 2025). Observations of CNC laser and fabrication lines with technical supervisors.
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- Kalpakjian, S. & Schmid, S.R. (2022). Manufacturing Engineering and Technology (8th Ed.). Pearson.
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Data Tabulation:

Table 1 — Sands Precision Pvt. Ltd.

Category	Parameter	Details
Organisation	Ownership	Private company — started by 5 partners, currently run by 2; met Mr. Dinesh Karanth (co-owner)
	Total Workforce	~30 employees
	Workforce Split	8 engineers and admin staff; remaining ~22 are shop-floor workers
	Shifts	1 shift per day
	Annual Output	~100 CNC machines per year
	Product	CNC turning centres and vertical machining centres (VMCs)
	Customers	Automobile sector; precision manufacturing industries across India
Process & Technology	Manufacturing Sequence	Design (legacy in-house design) → Procurement of parts from vendors → Assembly → Testing → Dispatch
	Machines/Equipment	Vertical milling machines, CNC turning centres, overhead crane for heavy component handling
	CAD/CAM Software	Third-party software from major vendors (Fanuc, Siemens); no proprietary software developed in-house

	NC Programming	G-code generated via imported CNC controllers fitted into their machines
	Assembly Method	Fully manual assembly; large parts lifted by crane, fitted and tightened manually by operators
Productivity & Quality	Cycle Time	15–20 days per CNC machine
	Quality Inspection	Dedicated inspection personnel; some measuring instruments used; no CMM mentioned
	Defect Rate	Not disclosed
	Machine Utilisation	~45% (batch production model — not continuous)
Automation	Automation Level	Low — assembly is entirely manual; only material handling (crane) is mechanised
	Control Systems	Fanuc, Siemens, and other major CNC controller brands integrated into their machines
	Robots/AGVs	None
Challenges	Primary Challenge	Cost — keeping production cost low is the biggest constraint
	Secondary Challenge	Skilled labour availability for precision assembly work
Market & Trends	Sectors Served	Automobile, aerospace, precision engineering
	Competitive Position	Assembler and integrator model — sources parts, adds design value, sells complete CNC machines

Table 2 — United Concord Industries (Concord United Products Pvt. Ltd.)

Category	Parameter	Details
Organisation	Ownership	Private company
	Total Workforce	250+ employees across 3 units in Peenya
	Workforce Split	Majority are designers and shop-floor staff; admin present; exact split not recorded
	Shifts	3 shifts per day — continuous production
	Facilities	3 manufacturing units in Peenya Industrial Area
	Product	Pelletizing equipment — machines that convert molten plastic/polymer into uniform pellets (granules) used as raw material for plastic product manufacturing
	Customers	Plastics industry, food processing, chemical industry; expanding into sustainable/recycling sector
Process & Technology	Manufacturing Sequence	Design → Raw material procurement → Cutting → Machining (finishing, blades, sharp-edge components) → Assembly → Testing → Dispatch
	Machines/Equipment	Electric arc welding machines, Electric Discharge Machining (EDM), CNC machining centres, automated cutting machines
	CAD/CAM Software	AutoCAD, SolidWorks, and additional advanced CAD/CAM platforms
	Design	Fully in-house design and engineering team

Productivity & Quality	Cycle Time	~6–7 days per unit (varies by order size and complexity)
	Quality Inspection	Precision measurement and functional testing conducted before dispatch
	Defect Rate	Not disclosed
	Machine Utilisation	~85% — continuous, high-volume production
Automation	Automation Level	Medium-High — most machining and cutting operations done by machines; human operators required for control, supervision, and assembly
	Robots/AGVs	None observed
	Control Systems	Not recorded; advanced CNC and automated systems in use
Challenges	Primary Challenge	No specific challenges mentioned
	Strategic Direction	Actively expanding into sustainable manufacturing and plastic recycling industry
Market & Trends	Sectors Served	Plastics, food processing, chemicals; growing focus on sustainability and recycling
	Competitive Position	Established pelletizing equipment manufacturer with in-house design capability and multi-unit production capacity

Table 3 — Sree Ranga Engineering Works

Category	Parameter	Details
Organisation	Ownership	Private company
	Total Workforce	~50 employees
	Workforce Split	Operators, technicians, and admin; exact split not recorded
	Shifts	3 shifts per day
	Business Model	Mixed — both job-shop (custom orders) and standard products
	Product/Service	Sheet metal laser cutting, bending, and fabrication for multiple industries
Process & Technology	Customers	Multiple sectors — no single dominant customer base
	Manufacturing Sequence	Design (in-house or as per customer requirement) → Laser cutting → Bending (if required) → Dispatch
	Key Machines	2 × CNC laser cutting machines (Chinese origin, ₹1.5 Cr each) — brands R3 and HSG
	CAD Software	AutoCAD, SolidWorks
	NC Program	Generated in-house or provided by client as per requirement
	Sheet Processed	Various sheet metal grades and thicknesses

Productivity & Quality	Cycle Time	~15 minutes per sheet (variable — order quantities range from 1 sheet to 100+ sheets)
	Quality Inspection	Minimal — laser cutting is inherently precise; no dedicated inspection stage observed
	Defect Rate	Not disclosed; laser precision reduces typical cutting defects
	Machine Utilisation	High — 3-shift operation indicates near-continuous machine use
Automation	Automation Level	Low-Medium — laser cutting is CNC-controlled but machine operation (loading, unloading, bending) is manual
	Control Systems	R3 and HSG CNC laser controller systems (observed on machines)
	Robots/AGVs	None
Challenges	Primary Challenge	None specifically mentioned
	Future Plans	None specifically mentioned
Market & Trends	Sectors Served	Diverse industrial sectors — sheet metal components for various applications
	Competitive Position	High-value capital equipment (₹1.5 Cr machines) enables precision cutting competitive with larger fabricators

Parameter	Sands Precision Pvt. Ltd.	United Concord Industries	Sree Ranga Engineering Works
Industry Type	CNC Machine Manufacturing	Pelletizing Equipment Manufacturing	Sheet Metal Fabrication & Laser Cutting
Ownership	Private — 2 active partners	Private	Private
Workforce	~30 employees	250+ employees	~50 employees
Shifts	1 shift/day	3 shifts/day	3 shifts/day
Product	CNC turning centres & VMCs	Pelletizing machines for plastic/polymer processing	Custom & standard sheet metal components
Key Equipment	VMCs, CNC turning centres, overhead crane	EDM, arc welding, CNC machining centres	2 × CNC laser cutters — R3 & HSG (₹1.5 Cr each)
CAD/CAM Software	Fanuc/Siemens (vendor-supplied)	AutoCAD, SolidWorks + advanced CAM	AutoCAD, SolidWorks
Manufacturing Sequence	Design → Procure → Assemble → Test → Dispatch	Design → Cut → Machine → Assemble → Test → Dispatch	Design → Laser Cut → Bend → Dispatch
Cycle Time	15–20 days / machine	6–7 days / unit	~15 min / sheet
Machine Utilisation	~45%	~85%	High (near-continuous)
% Automation	~20% — crane only; full manual assembly	~65% — machining automated; assembly manual	~40% — cutting automated; bending & handling manual
Quality Inspection	Dedicated staff + measuring instruments	Precision measurement + functional testing	Minimal — laser precision inherently accurate
Key Challenge	Cost reduction + skilled labour	Expanding into recycling/sustainability sector	None mentioned

